SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE,UNLIMITED POTENTIAL

School of Engineering and Technology

Internal Assessment - I (IA-1)

Department: Physics
Course Name: Engineering Physics
Batch: 2025-26
Semester: II

Date: 14/05/2026
Course Code: 25BEPHY202
Max. Marks: 50
Duration: 90 Mins

Q No	Question	Marks	RBT Level	CO
1.	a. Mention the properties of wave function.	4	L2	CO1
	b. Discuss the working of Controlled Z gate mentioning its matrix representation and truth table.	6	L2	CO2
	OR			
2.	a. Calculate the de Broglie wave length associated with an electron accelerated through a potential difference 0f 2kV (Given: $h = 6.625x10^{-34}$ Js, $m = 9.1x10^{-31}$kg). 2.75×10^{-11} m	5	L3	CO1
	b. Give the matrix representation of Single qubit and Multiple qubits.	5	L2	CO2
3.	a. Develop Schrödinger's time-independent equation in one dimension.	10	L2	CO1
	b. Derive an expression for the de Broglie wavelength in terms of kinetic energy and accelerating potential.	10	L2	CO1
	OR			
4.	a. Using the Schrödinger wave equation, derive the expression for the energy Eigen value and the Eigen functions of a particle in a one-dimensional box.	10	L2	CO1
	b. State and explain Heisenberg's uncertainty principle.	5	L2	CO1
	c. An electron is bound in an one dimensional potential well of width 1Å, but infinite wall height. Calculate its energy value in the ground state and second excited state. (Given:$h = 6.625x10^{-34}$ Js, $m = 9.1x10^{-31}$kg) $n=1$ 37.7 eV $n=3$ 339eV	5	L3	CO1
5.	a. Give the definition, matrix form, operations and truth table for Pauli's X gate, Y gate and Z gate.	10	L2	CO2
	b. What are Qubits? Explain any four properties of Qubit.	5	L2	CO2
	c. Verify the given matrix is unitary or not $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}$	5	L3	CO2
	OR			
6.	a. What is Controlled NOT Gate? Give its matrix form and truth table. Explain the operation of CNOT gate on $\|0\rangle$ and $\|1\rangle$ state.	9	L2	CO2
	b. Explain how Hadamard gate transforms $\|0\rangle$ and $\|1\rangle$ state.	6	L2	CO2
	c. State and explain Moore's law and discuss its end with graph.	5	L2	CO2

Course outcomes: Students will be able to
CO1: Use the quantum mechanical laws of physics and solve basic problems.
CO2: Explore the applications of quantum mechanics in quantum computing.

$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

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SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE,UNLIMITED POTENTIAL

School of Engineering and Technology

SRN: | | | | | | | | | | | | | |

Internal Assessment – 1 (CIA-1)

Department: Physics | **Date:** 09/12/2025

Course Name: Engineering Physics | **Course Code:** 25BEPHY102

Batch: 2025-26 | **Max. Marks:** 50

Semester: I | **Duration:** 90 Mins

Note: Answer all the questions. Q1 & Q2 carries 10 Marks each, Q3 - Q6 carry 20 Marks each

Q No	Question	Marks	RBT Level	CO
1.	a. What is a matter wave? Explain the de Broglie wavelength.	5	1	1
	b. What are qubits? List five properties of qubits.	5	1	2
	OR			
2.	a. State and explain Heisenberg's uncertainty principle.	5	2	1
	b. What is a unitary matrix? Give an example.	5	2	2
3.	a. Derive an expression for the de Broglie wavelength in terms of kinetic energy. Determine the wavelength of an electron accelerated by a potential V	10	3	1
	b. Develop Schrödinger's time-independent equation for one dimension.	10	2	
	OR			
4.	a. Using the Schrödinger wave equation, derive the expression for the energy eigenvalues for the first three energy states and the Eigen functions of a particle in a one-dimensional box.	10	3	1
	b. An electron has a speed of 4.8 x 10^5 m/s accurate to 0.012%. With what accuracy can the position of electron be located?	5	3	
	c. An electron is bound in 1-D potential well of width 0.18nm, find its energy value in eV in the second excited state.	5	3	
5.	a. State and explain Moore's law with graph and discuss its failure.	5	2	2
	b. Give the Definition, matrix form, operations and truth table for. Pauli's X gate, Y gate and Z gate.	10	2	
	c. Verify the given matrix is unitary or not $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	5	3	
	OR			
6.	a. Explain the representation of Qubits. Give the matrix representation for single qubit and 2 qubits.	5	1	2
	b. Give the symbol, truth table, and matrix for the controlled-NOT (CNOT) gate. Explain the operation for each possible input.	10	2	
	c. Define the transpose and the conjugate of a matrix, with an example for each.	5	1	

Course outcomes: Students will be able to

CO1: Use the quantum mechanical laws of physics and solve basic problems.

CO2: Explore the applications of quantum mechanics in quantum computing.

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